Year II — Algebra

Based on lectures by Various Lecturers

Notes taken by Anonymous

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Chapter 1

Rings, subrings and fields

1.1 Definition of a Ring

Before being able to define a ring, we must first redefine *binary operations*:

Definition 1.1.1 (Binary Operation)**.** A *binary opeation* is a function that takes two elements as an argument and produces a single new output value. Examples include addition, multiplication and subtraction.

Remark 1.1.2. All rings form Abelian groups (review first year if unsure what this means)

Definition 1.1.3 (Ring)**.** A ring R is a set together with two binary operations (usually, but not always, written,$+\cdot$) called "addition" and "multiplication", such that $+$ makes R into an abelian group, and such that the following conditions hold:

i) **Existence of an Identity** - There exists an element $1 \in R$ such that $1 \cdot x = x \cdot 1 = x$, for all $x \in R$.

ii) **Associativity** - For any $x, y, z \in R$ we have $x(yz) = (xy)z$.

iii) **Distributivity** - For any $x, y, z \in R$ we have $x(y + z) = xy + xz$ and $(y + z)x = yx + zx$.

1.2 Examples of Rings

– Let V be a finite dimensional vector space over $\mathbb{C}$. Let

$$\mathrm{End}(V) = \{f : V \to V \mid f \text{ is a linear map } \}.$$

An element of $\mathrm{End}(V)$ is usually called a (linear) endomorphism. We can add two such functions like we have seen before:

$$(f + g)(v) = f(v) + g(v).$$

Now we can't directly "multiply" two endomorphisms, because V does not have any multiplication (V is not in general a ring). However, we can compose two endomorphisms

$$(f \circ g)(v) = f(g(v)),$$

and composition is a binary operation on V. From Linear Algebra I, we already know that $\mathrm{End}(V)$ is an abelian group under addition. To check that this is a ring, we need to check the three properties:

Identity element is provided by the identity map $Id : V \to V$. Associativity and distributivity are also not hard to check. For example we need to establish that if $f, g, h \in \text{End}(V)$ we have

$$f \circ (g \circ h) = (f \circ g) \circ h.$$

That is, for every vector $v \in V$ we have that $(f \circ (g \circ h))(v) = ((f \circ g) \circ h)(v)$. For the definition of the "multiplication" we have that the left hand side is equal to

$$(f \circ (g \circ h))(v) = f((g \circ h)(v)) = f(g(h(v))$$

and

$$((f \circ g) \circ h)(v) = (f \circ g)(h(v)) = f(g(h(v))$$

That is the two endomorphisms are having the same effect on any vector v of V. That is they are the sane maps. Similarly one checks distributivity.

- $\mathbb{N}$ is not a ring ($1 - 2 \notin \mathbb{N}$)
- $\mathbb{Z}$ is a ring but not a field: $\frac{1}{2} \notin \mathbb{Z}$.
- $\mathbb{Q}[x]$ is a ring but not a field: $\frac{1}{x} \notin \mathbb{Q}[x]$.
- $\mathrm{M}_n(\mathbb{Q})$ is a ring but not a field: there is no way we can make sense of division. For example, $\frac{1}{\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}}$ is meaningless since that matrix does not have an inverse.
- The set $C(\mathbb{R})$ is a ring but not a field: the constant function $1 : \mathbb{R} \to \mathbb{R}$ belongs to the set, and so does the identity function Id defined by $\text{Id}(x) = x$ for all $x \in \mathbb{R}$. However, the function $\frac{1}{\sqrt{d}}$ is not continuous at $x = 0$, so does not belong to $C(\mathbb{R})$.

Method 1.2.1 (Determining a Ring)**.** To determine a ring, one should first check that the ring forms an abelian group with respect to the 'addition' operation and that the following requirements hold for the 'multiplication' operation:

i) **Existence of an Identity** - There exists an element $1 \in R$ such that $1 \cdot x = x \cdot 1 = x$, for all $x \in R$.

ii) **Associativity** - For any $x, y, z \in R$ we have $x(yz) = (xy)z$.

iii) **Distributivity** - For any $x, y, z \in R$ we have $x(y + z) = xy + xz$ and $(y + z)x = yx + zx$.

1.3 The integers modulo n

Definition 1.3.1 (Equivalence Relation)**.** A (binary) relation $\sim$, on a set S is said to be an equivalence relation, if and only if it is reflexive, symmetric and transitive. That is, for all a, b, and c in the set S we have:

i) $a \sim a$ (reflexivity).

ii) $a \sim b$ if and only if $b \sim a$ (symmetry).

iii) If $a \sim b$ and $b \sim c$ then $a \sim c$ (transitivity).

Definition 1.3.2 (Equivalence Class)**.** The equivalence class of a under the given relation $\sim$, denoted by $[a]$ (or $\bar{a}$) is defined as

$$[a] = \{x \in S : x \sim a\}.$$

Note that for two $a, b \in S$ with $a \sim b$ we have $[a] = [b]$. In particular when we write $[a]$ we call a the representative of the class $[a]$. Clearly an equivalence class may have more than one representative.

From these structures we can construct another set:

$$S/\sim := \{[a] : a \in S\}$$

That is, this is the set of equivalence classes on the set S with respect to the given equivalence relation $\sim$.

Definition 1.3.3 (Integers Modulon n)**.** The integers modulo n. Let $n \in \mathbb{N}$ be a natural number. Let us consider the set of integer numbers $\mathbb{Z}$ and define the following relation: two integers a and b are equivalent (i.e. $a \sim b$ in the above notation) if n divides $a - b$. The most common notation for this is $a \equiv b \pmod{n}$.

Proof of 'the relation is an equivalence relation'.

- it is reflexive as $a - a = 0$ and hence divisible by n,
- it is symmetric as n divides $a - b$ if and only if n divides $b - a = -(a - b)$.
- And finally for the last condition we consider $a, b, c \in \mathbb{Z}$ and further assume that n divides $a - b$ and $b - c$, that is $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$. That is, $a - b = kn$ and $b - c = rn$ for some $k, r \in \mathbb{Z}$. But then $a - c = a - b + b - c = (k + r)n$ and hence n divides $a - c$ i.e. $a \equiv c \pmod{n}$.

□

Definition 1.3.4 (Residue Class)**.** For an integer a we denote by $\bar{a}$ (this is the most commonly used notation) the equivalence class of a, that is

$$\bar{a} = \{x \in \mathbb{Z} : n \text{ divides } x - a\} = \{x \in \mathbb{Z} : x = a \pmod{n}\}$$

Now note that when we divide an integer by n we will always get a remainder among the numbers

$$0, 1, , \ldots, n - 1.$$

That is, every integer belongs to one of the following equivalence classes:

$$\{\overline{0}, \overline{1}, \ldots, \overline{n-1}\},$$

We will denote this set (of equivalence classes) by $\mathbb{Z}/n$. The elements of $\mathbb{Z}/n$ are called residue classes.

Remark 1.3.5. Make sure that if you ever write $\mathbb{Z}_n$ that you explain that you mean $\mathbb{Z}/n$ as it can cause confusion.

1.4 Subrings and Fields

Definition 1.4.1 (Subring)**.** A *subring* S of a ring R is a set $S \subseteq R$ such that

i) $0 \in S, 1 \in S$ where 0 is the 'additive' identity and 1 it the 'multiplicative identity',

ii) $a, b \in S \Rightarrow a + b \in S$,

iii) $a, b \in S \Rightarrow ab \in S$,

iv) $a \in S \Rightarrow -a \in S$.

Remark 1.4.2 (Intuitive definition of Subring)**.** We can shorten this into an informal definition of 'A subring contains 0_R, 1_R and is closed under addition, subtraction and multiplication.

Remark 1.4.3 (Subrings are Rings)**.** A subring of any ring is a ring itself and is equipped with addition and multiplication of the original ring.

Method 1.4.4 (Showing a Subring)**.** We can quickly show a subring by showing:

- the additive and multiplicative identities exist in the set,
- it's closed under addition
- it's closed under multiplication
- it contains additive inverses.

Method 1.4.5 (Determining a Ring quickly)**.** Rather than displaying a set satisfies all of the properties of a ring, it is sufficient to show that the set is a subring of a ring. It is therefore a ring itself.

Definition 1.4.6 (Field)**.** A ring R is called a field if:

i) R is commutative,

ii) $1 \neq 0$ in R (i.e., R has at least two elements),

iii) For any $a \in R$ such that $a \neq 0$, there exists a $b \in R$ such that $ab = 1$ (b is called the inverse of a).

Chapter 2

Arithmetic in $\mathbb{Z}$ and $F[x]$

2.1 Divisibility in $\mathbb{Z}$

Definition 2.1.1 (Division)**.** In a commutative ring R, an element $a \in R$ is said to divide an element $b \in R$, written $a \mid b$, if there exists an $x \in R$ such that

$$b = ax.$$

We will sometimes write $a \nmid b$ if a does not divide b.

Remark 2.1.2. It is easy to show from the definition that if $a \mid b$ and $a \mid c$ for $a, b, c \in R$, then $a \mid (bx+cy)$, for any $x, y \in R$. In particular, if a divides two elements, then it divides the sum and difference of those two elements.

Lemma 2.1.3 (Well-Ordering Principle (WOP))**.** Let S be a non-empty subset of $\mathbb{N}_0 = \mathbb{N} \cup \{0\}$. Then S has a smallest element.

Proposition 2.1.4. Let $a \in \mathbb{Z}, b \neq 0$. Then there exist in tegers q and r such that

$$a = q \cdot b + r, \quad \text{and} \quad 0 \leq r < |b|.$$

(q is called the quotient and r the remainder).

Proof. Insert proof here □

Example 2.1.5. Find the integer quotient and remainder of 435 divided by 43 in $\mathbb{Z}$. Long division with remainder gives:

$$435 = 43 \cdot 10 + 5,$$

and $5 < 43$, so we stop. So the integer quotient is $q = 10$ and the remainder is $r = 5$.

Theorem 2.1.6 (Euclidean Algorithm in $\mathbb{Z}$)**.** Let $a, b \in \mathbb{Z}$, not both zero. Then $\gcd(a, b)$ exists and can be computed by the Euclidean algorithm (given in the proof). Moreover, if $d = \gcd(a, b)$, then there exist $x, y \in \mathbb{Z}$ such that

$$ax + by = d.$$

Proof. Insert proof here □

Example 2.1.7. Example 2.5. We use the Euclidean Algorithm to compute $d = \gcd(455, 1235)$ and find integers x, y such that $d = 455x + 1235y$.

Solution: Division with remainder:

$$\begin{aligned} 1235 &= 2 \cdot 455 + 325 \\ 455 &= 1 \cdot 325 + 130 \\ 325 &= 2 \cdot 130 + 65 \\ 130 &= 2 \cdot 65 + 0. \end{aligned}$$

When the remainder is 0 , we stop. The gcd will be the last non-zero remainder, here $d = 65$. Solving for the remainders, substituting back and simplifying, we get

$$\begin{aligned} 65 &= 325 - 2 \cdot 130 \\ &= 325 - 2(455 - 1 \cdot 325) \\ &= \cdots \\ &= 455 \cdot (-8) + 1235 \cdot 3. \end{aligned}$$

so $x = -8$ and $y = 3$.

Lemma 2.1.8 (Primes that divide products divide at least one factor)**.** If p is a prime and $p \mid ab$ for some $a, b \in \mathbb{Z}$, then $p \mid a$ or $p \mid b$. (In other words, p prime $\Rightarrow (\forall a, b \in \mathbb{Z}, p|ab \Rightarrow p|a$ or $p \mid b)$.)

Theorem 2.1.9 (Fundamental Theorem of Arithmetic)**.** Let $n \in \mathbb{N}, n > 1$. Then n can be written as a product of primes in a unique way (up to the order of the factors). More precisely, there are distinct primes $p_1, \ldots, p_r$ and $e_1, \ldots, e_r \in \mathbb{N}$ such that

$$n = p_1^{e_1} \cdots p_r^{e_r},$$

and if $n = q_1^{f_1} \cdots q_s^{f_s}$ for some distinct primes $q_1, \ldots, q_s$ and $f_1, \ldots, f_s \in \mathbb{N}$, then $r = s$, and up to re-ordering, $p_i = q_i$ and $e_i = f_i$ for all i.

2.2 Divisibility in $F[x]$

Definition 2.2.1 (Degree)**.** For $f(x) = a_0 + a_1 x + \cdots + a_n x^n \in F[x]$, its *degree* is

$$\deg f := \begin{cases} \max\{i \mid a_i \neq 0\} & \text{if } f(x) \neq 0, \\ -\infty & \text{if } f(x) = 0. \end{cases}$$

Then we have the following basic properties for all $f(x), g(x) \in F[x]$:

- $\deg(fg) = \deg f + \deg g$,
- $\deg(f + g) \leq \max\{\deg f, \deg g\}$. Note that if $f(x) \neq g(x)$, then equality holds

Remark 2.2.2. We define $\deg(f) = -\infty$ if $f(x) = 0$ to preserve the properties of degree in the definition.

If $f = 0$ and $\deg 0 = \deg(0 \cdot g) = \deg 0 + \deg g$, then we can't allow $\deg 0$ to be in $\mathbb{Z}$, but it would work if $\deg 0 = \pm\infty$. However, if also $\deg g = \deg(0 + g) \leq \max\{\deg 0, \deg g\}$, for all g, then we can't allow $\deg 0 = \infty$, but it does work if we set $\deg 0 = -\infty$.

Proposition 2.2.3. Let $f(x), g(x) \in F[x]$ with $g(x) \neq 0$. Then there exist polynomials $q(x), r(x) \in F[x]$ with $\deg r < \deg g$ such that

$$f(x) = q(x)g(x) + r(x).$$

Example 2.2.4 (Long division of $f(x) = x^3 + x^2 - 3x - 3$ by $g(x) = x^2 + 3x + 2$, with remainder)**.**

term to get rid of		$x - 2$
x^3	$x^2 + 3x + 2$	$x^3 + x^2 - 3x - 3$ $-\left(x^3 + 3x^2 + 2x\right)$
$-2x^2$		$-2x^2 - \left(-2x^2 - 6x - 3 \ -7\right)$
		$x + 1$

We have obtained a remainder of degree 1 , which is smaller than $\deg g$, so we stop:

$$f(x) = (x - 2)g(x) + x + 1.$$

Thus $q(x) = x - 2$ and $r(x) = x + 1$. Think of $q(x)$ as the "quotient" and $r(x)$ as the "remainder".

Proof. insert proof here □

2.3 GCD's and the Euclidean Algorithm in $F[x]$

Definition 2.3.1 (Monic Polynomial)**.** A polynomial in $F[x]$ is said to be monic if its leading coefficient is 1 .

Definition 2.3.2 (GCD for Polynomials)**.** Let $f(x), g(x) \in F[x]$ with either $f(x) \neq 0$ or $g(x) \neq 0$. A greatest common divisor (GCD) of $f(x)$ and $g(x)$ is a polynomial of highest degree that divides both $f(x)$ and $g(x)$.

Example 2.3.3. Example 2.11. Let $f(x) = x^2 + 1$ and $g(x) = x^2 + 3x + 1$ in $\mathbb{Q}[x]$. We perform the Euclidean algorithm to find a $\gcd(f(x), g(x))$. Long division of $f(x)$ by $g(x)$:

	1	
$x^2 + 3x + 1$	$x^2 + 1 - \left(x^2 + 3x + 1\right)$	so $f(x) = g(x) - 3x$
	$-3x$	

Next, divide the divisor $g(x)$ by the remainder $-3x$:

	$-\frac{1}{3}x + 1$	
$-3x$	$x^2 + 3x + 1$ $-\left(x^2 + 1\right)$	so $g(x) = \left(-\frac{1}{3}x + 1\right)(-3x) + 1$
	1	

‘ Next, divide the divisor $-3x$ from the previous step by the remainder 1: This obviously just gives

$$-3x = (-3x) \times 1 + 0.$$

Now the remainder is 0 , so we stop. A GCD of $f(x)$ and $g(x)$ is given by the last non-zero remainder: 1.

Important for later: We can use this to write 1 as a linear combination of $f(x)$ and $g(x)$:

$$\begin{aligned} 1 &= g(x) - \left(-\frac{1}{3}x - 1\right)(-3x) \\ &= g(x) + \left(\frac{1}{3}x + 1\right)(f(x) - g(x)) \\ &= \left(\frac{1}{3}x + 1\right) f(x) - \frac{1}{3}xg(x). \end{aligned}$$

Definition 2.3.4 (Degree Function)**.** Define the degree function $d : R\backslash\{0\} \to \mathbb{N}$ by

$$d(a) := \begin{cases} |a| & \text{if } a \in \mathbb{Z}, \\ \deg a & \text{if } a \in F[x]. \end{cases}$$

For $0 \in \mathbb{Z}, |0| = 0 \notin \mathbb{N}$ and for $0 \in F[x], \deg 0 = -\infty \notin \mathbb{N}$, which is why the domain of the function d is $R\backslash\{0\}$, rather than R

Remark 2.3.6. The use of the degree function is to generalise the definition of the Euclidean algorithm for both polynomials and integer rings. This is to carefully define the inequalitiy in the definition.

Lemma 2.3.7. Let R be either $\mathbb{Z}$ or $F[x]$. If $a, b \in R$ are such that $a \mid b$, then $d(a) \leq d(b)$.

Proof. Suppose that $a \mid b$, that is, $b = am$, for some $m \in R$. Then $d(b) = d(am) = |a| \cdot |m| \geq |a|$ when $R = \mathbb{Z}$ and $d(b) = d(am) = \deg a + \deg(m) \geq \deg a = d(a)$ when $R = F[x]$. □

Theorem 2.3.8 (Euclidean Algorithm - More Generally)**.** Let R be either $\mathbb{Z}$ or $F[x], F$ a field and let $a, b \in R$ such that $a \neq 0$ or $b \neq 0$. Then

i) A GCD of a and b exists and can be computed by the Euclidean algorithm (given in the proof);

ii) If $R = F[x]$, the monic GCD is unique and we denote it by $\gcd(a, b)$;

iii) If d is a GCD of a and b, then there exist $\alpha, \beta \in R$ such that $\alpha a + \beta b = d$.

Proof. insert proof here □

2.4 Units and Irreducibles, unique factorisation

Definition 2.4.1 (Invertible / Unit)**.** Let R be a ring. An element $a \in R$ is said to be invertible or a unit in R if there exists a $b \in R$ such that $ab = ba = 1$ (b is then the inverse of a and we write $b = a^{-1}$). The set of units in R is denoted $R^\times$.

Example 2.4.2.

- If a is a unit, then its inverse is unique: $ab = 1 = ac = ca \Rightarrow cab = c \Rightarrow b = c$.
- The units in $\mathbb{Z}$ are ± 1 : if $a \in \mathbb{Z}$ is a unit and $ab = 1$, then $a \mid 1$.
- The units in $F[x], F$ a field, are just the units in F, that is, the non-zero constants. Proof: $a(x) \in F[x]$ is a unit with $a(x)b(x) = 1$, for some $b(x) \in F[x]$, then $\deg a + \deg b = \deg 1 = 0$, and since $\deg a, \deg b \geq 0$, we must have $\deg a = \deg b = 0$, which means that a (and b) are non-zero constant polynomials.

Definition 2.4.3 (Irreducible Polynomial)**.** A polynomial $f(x) \in F[x]$ is irreducible if it is non-constant and does not have a proper factorisation, that is, $f(x)$ cannot be written as $f(x) = g(x)h(x)$ with $\deg g > 0$ and $\deg h > 0$. We generalise this definition to one that also captures the primes in $\mathbb{Z}$.

Definition 2.4.4 (Irreducible Element of a Ring)**.** Let R be a ring. An element $r \in R$ is called irreducible if is satisfies

- r is not a unit;
- If $r = ab$, for some $a, b \in R$, then a or b is a unit.

Lemma 2.4.5 (Generalised prime statement)**.** Let R be either $\mathbb{Z}$ or $F[x], F$ a field. If $p \in R$ is irreducible and $p \mid$ ab for some $a, b \in R$, then either $p \mid a$ or $p \mid b$.

Proof. Insert proof here □

Theorem 2.4.6. Let R be either $\mathbb{Z}$ or $F[x], F$ a field and let $a \in R$ be a non-zero non-unit. Then a can be factorised uniquely into a product of irreducible elements, up to order of the factors and multiplication by invertible elements.

Proof. Insert proof here □

Chapter 3

Units, zero divisors and integral domains

3.1 Examples of units in rings

We have seen in the previous section that the units in $\mathbb{Z}$ are ± 1 and the units in $F[x], F$ a field are the non-zero constants, that is, $F^\times$. We now look at examples of units in other rings. If R is a ring the units $R^\times$actually form a group under multiplication in R :

$$a, b \in R^\times \implies ab \in R^\times$$

because if $c, d \in R$ are elements such that $ac = 1$ and $bd = 1$, then $(ab)(dc) = 1$, so ab is a unit. Moreover, we have an identity element 1 in the group, the multiplication is associative (because it is so in the ring R), and every element in $R^\times$has an inverse by definition. We will however not need this fact in the ring theory part.

Example 3.1.1. If $R = \mathrm{M}_n(F)$ (ring of matrices over a field F), then $R^\times = \{x \in \mathrm{M}_n(F) \mid \det x \neq 0\}$ (by Lin Alg. I). This group is denoted $\mathrm{GL}_n(F)$ (the General Linear group over F).

Proposition 3.1.2. An element $\bar{a} \in \mathbb{Z}/n$ is a unit iff $\gcd(a, n) = 1$

Proof. Insert Proof Here □

Corollary 3.1.3. $\mathbb{Z}/p$ is a field if and only if p is a prime.

Proof. insert proof here □

3.2 Zero Divisors and Integral Domains

Definition 3.2.1 (Zero Divisor)**.** Let R be a ring. If $a \in R$ is such that $ab = 0$ for some $b \neq 0$, then a is called a zero divisor.

Definition 3.2.2 (Integral Domain)**.** Definition 3.5. A ring R which is commutative, has at least two elements $(0 \neq 1)$, and does not have any zero divisors apart from 0 , is called an integral domain.

The essential property of an integral domain is the multiplicative cancellation property:

$$ax = bx \quad \text{and } x \neq 0 \implies a = b$$

and this holds even if x is not a unit and you can't divide by it.

Lemma 3.2.3. $\mathbb{Z}/n$ is an integral domain if and only if n is a prime.

Proof. Insert Proof here □

Proposition 3.2.4. If R is an integral domain (e.g., a field), then the ring of polynomials $R[x] = \{a_0 + a_1x + \cdots + a_nx^n \mid a_i \in R\}$ is also an integral domain.

Proof. Insert proof here □